BERT Sentiment Analysis Project Summary- 28/10/25(Tuesday):

Project Goal: Implement a BERT-based sentiment analysis model to classify text as positive (label 1) or negative (label 0) and evaluate its performance on a small, custom dataset. This project fulfills the M.Tech internship requirement for [Insert Internship Company Name].

Model: BERT-base-uncased (Bidirectional Encoder Representations from Transformers) Framework: Hugging Face transformers and PyTorch Hardware: NVIDIA T4 GPU (Google Colab)

☑ Training and Evaluation Results

The model was fine-tuned for 3 epochs on a subset of the IMDb dataset (2,000 training samples, 500 test samples). The T4 GPU significantly accelerated the training process.

Final Test Metrics

These metrics were calculated on the unseen 500-sample test subset after the 3rd epoch.

Metric	Score	Interpretation
Accuracy	0.9080 (90.80%)	The overall percentage of correct predictions.
F1 Score	0.905	The harmonic mean of Precision and Recall, indicating strong, balanced performance.
Precision	0.9202	Of all reviews predicted as Positive, 92.02% were correct.
Recall	0.8902	Of all truly Positive reviews, 89.02% were correctly identified.
Total Training Time	≈ 10 minutes	Time for the entire fine-tuning process.

Custom Dataset Testing

The model was tested against a small set of manually created sentences to verify real-world application.

Input Sentence	Predicted Sentiment	Confidence Score	Notes
"The movie was absolutely mesmerizing and the acting was top-notch!"	Positive (LABEL_1)	0.9677	High confidence; successfully classified strong positive language.

"I would not recommend this product; it broke after only one use."	Negative (LABEL_0)	0.8742	Correctly identified negative intent via explicit non-recommendation.
"It was neither good nor bad, just utterly forgettable."	Negative (LABEL_0)	0.5203	Low Confidence. The sentence is highly neutral/ambiguous. The model leaned slightly towards Negative but expressed uncertainty, which is a good indicator of handling mixed input.
"The presentation of the food was disgusting."	Negative (LABEL_0)	0.8901	High confidence; correctly identified strong negative language.

Code Summary and Key Steps

The solution uses the Hugging Face ecosystem for efficiency:

Step	Description	Key Code Components
Data Preparation	Loaded the IMDb dataset, tokenized text using tokenizer.encode_plus with a max length of 512, and selected a small subset for quick training.	datasets.load_dataset("imdb"), AutoTokenizer, .map(tokenize_function)
Model Fine-Tuning	Loaded the BertForSequenceClassification model and used the high-level Trainer API for optimization and metric calculation. W&B tracking was enabled/disabled (as per choice).	BertForSequenceClassification.from_pretrained(), TrainingArguments, Trainer.train()
Evaluation	Defined a custom compute_metrics function using sklearn to calculate and report Accuracy, Precision, Recall, and F1.	precision_recall_fscore_support, accuracy_score
Testing/Deployment	Saved the fine-tuned model and used the pipeline abstraction to load and run inference on custom sentences easily.	model.save_pretrained(), transformers.pipeline()